

MiCA Compliance Copilot — Documentation

Course: AI for Developers — Building with Large Language Models (AUEB). **Type:** Generative-AI application — *not a chatbot*: AI logic embedded in a clean software architecture (FastAPI backend + functional UI + GenAI layer).

1. Title, description, purpose

MiCA Compliance Copilot is a Retrieval-Augmented + agentic assistant for the EU **Markets in Crypto-Assets Regulation** (Regulation (EU) 2023/1114, "MiCA").

Purpose. Answer MiCA questions *grounded in the actual regulation* with article-level citations, classify a token/service under MiCA, and look up real ESMA register data — while **refusing to answer** when the indexed corpus doesn't support a grounded answer. In a compliance context, a confident-but-wrong answer is worse than "I don't know," so retrieval + citation + abstention are the whole point.

2. Use scenario & functional requirements

Scenario. A compliance analyst or crypto founder asks, "*What backs the reserve of an asset-referenced token?*" or "*Is my euro stablecoin an EMT, and what must I do?*" They need an answer they can trust and trace back to a specific article — not a plausible paraphrase.

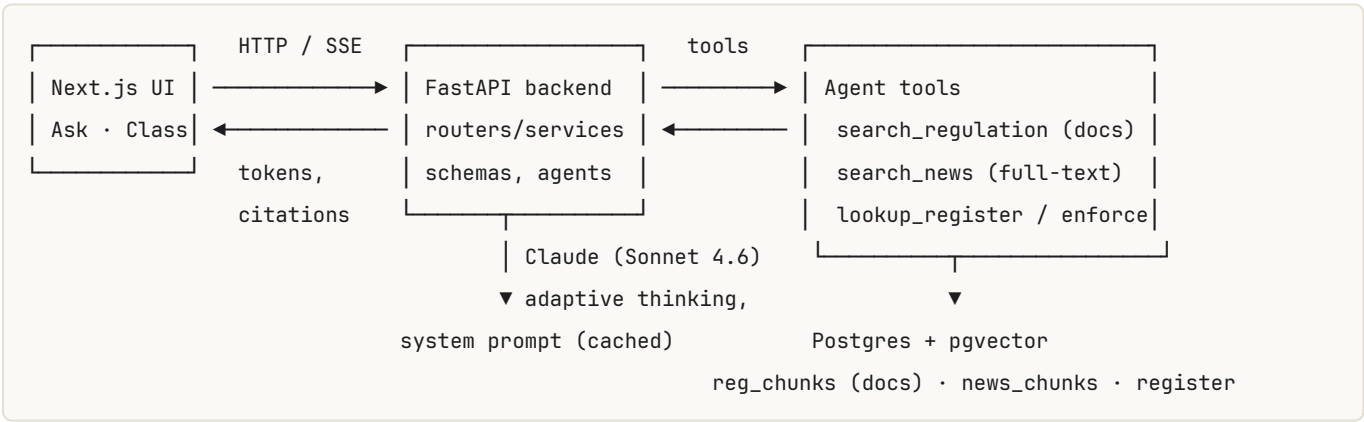
Functional requirements

- F1. Answer MiCA questions grounded in retrieved regulation text, with per-article citations.
- F2. Abstain (don't fabricate) when retrieval doesn't support an answer.
- F3. Classify a described token/service: asset type (ART / EMT / other / out-of-scope), the crypto-asset services (a–j) involved, and the resulting obligations — as structured data.
- F4. Look up whether a named entity is an authorised CASP / EMT issuer, or is flagged.
- F5. Stream answers and show the agent's tool-use trace; expose Swagger docs.

3. Technologies used & rationale

Layer	Choice	Why
Backend	FastAPI	Async, typed (pydantic) request/response, automatic Swagger/OpenAPI — required by the brief and a clean orchestration point.
LLM	Claude Sonnet 4.6 (agent loop, default) + Haiku 4.5 (cheap tasks/judge)	Sonnet balances tool-use reasoning and cost; AGENT_MODEL=c <code>claude-opus-4-8</code> is a one-line upgrade for maximum quality. Adaptive thinking + effort .
Vector store	Postgres 16 + pgvector	One self-contained DB (<code>docker compose up</code>) holds both the embeddings and the register snapshot; reproducible for a grader.
Embeddings	mxbai-embed-large-v1 (local, default) / Voyage voyage-law-2	local model is 1024-dim, ~0.64 GB, runs offline with no API key (reproducible key-free); <code>voyage-law-2</code> is the legal-tuned paid alternative.
Sources	EUR-Lex + ESMA/EBA (docs) · RSS (news) · ESMA register CSVs (registers)	<code>feedparser</code> + <code>trafilatura</code> + <code>pypdf</code> ingest the regulation, RTS/ITS, guidelines/Q&As, full news articles, and the real ESMA registers (incl. Title II white papers, whose token is read from each document).
UI	Next.js 15 / React 19	Streaming chat + structured result rendering; design system adapted from the author's <code>mica-dashboard</code> (warm-paper + EU-blue, hand-rolled — no Tailwind).

4. Architecture & data flow



Request flow for `/chat` : 1. UI POSTs the question; backend opens a streamed agent loop. 2. Claude routes: regulatory questions → `search_regulation` (embed → match `reg_chunks`); current-status questions → `search_news` (recency-ranked `news_chunks`) + `lookup_register / check_enforcement` . 3. Claude answers **only from retrieved sources** — citing documents by article/instrument and news by source + date — and abstains otherwise; the backend streams `token / tool / citations / done` .

Ingestion. `app.rag.docs_ingest` fetches the regulation + RTS/ITS (EUR-Lex) and ESMA/EBA guidelines/Q&As (PDF/HTML) → article/section chunks → `reg_chunks` (+ register seed). `app.news.poll` polls 14 RSS feeds, fetches **full article text** (`trafilatura`), triages relevance (Haiku), and embeds into `news_chunks` (recency-ranked). Both idempotent; the document corpus rebuilds offline from a committed cache.

5. FastAPI endpoints & services

Routers (`app/routers/`): `chat` (`/chat` SSE, `/chat/sync`), `classify` (`/classify`), `registry` (`/registry/search`), `health` (`/health`). All inputs/outputs are pydantic schemas (`app/schemas`), giving validation + the Swagger contract.

Services (`app/services/`): - `llm.py` — the Claude client, the **manual agentic tool loop**, **SSE streaming**, **prompt caching**, and **structured classification**. - `rag.py` — embed query → vector search → build the citable context block → dedup citations. - `registry.py` — read-only register lookups (degrade gracefully if a table is empty).

Agent layer (`app/agents/`): `tools.py` (tool definitions + local dispatcher), `prompts.py` (the cached system prompt + MiCA map + the classification JSON schema).

RAG layer (`app/rag/`): `embed.py` (Voyage/local behind one interface), `chunk.py` (article-aware chunking), `store.py` (pgvector upsert/search), `docs_ingest.py` (document corpus), `eurlex.py` (any-CELEX fetch/parse), `pdf.py` (PDF/HTML extraction).

News layer (`app/news/`): `sources.py` (14 RSS feeds), `poller.py` (conditional-GET, dedup, full-text fetch, classify, embed), `classify.py` (Haiku triage), `entities.py` (entity match), `store.py` (recency-weighted `news_chunks` search), `poll.py` / `scheduler.py` (CLI / APScheduler).

Register layer (`app/register/`): `sync.py` syncs the **real** ESMA register CSVs (CASPs, EMT/ART issuers, ~877 Title II white papers, warnings — `parse.py` handles dates/arrays/dedupe), and `whitepapers.py` **reads each white-paper document** (PDF/HTML) with Haiku to extract its token name + ticker (ESMA's register has no token-name field). `services/registry.py` then searches all of these by firm OR token, so "does Cardano/ADA have a MiCA white paper?" is answered from real data with the white-paper URL.

6. UI & interaction

- **Ask MiCA** (`/`): a streaming chat. Tool calls appear as chips ("Searched MiCA corpus ...", "Searched news ..."), the answer streams token-by-token, and a **Citations** panel groups sources into **Regulation & documents** (article/instrument + EUR-Lex/PDF link) and **News** (source + date + link). If the model abstains, a "No grounding — answer withheld" badge shows.
- **Classify** (`/classify`): paste a token/service description → a structured card with the asset type, the implied a–j services, the obligations (each tied to an article), and citations.

The UI is evaluated on functionality, clarity, and ease of use; example prompts are provided so a reviewer can exercise every path in one click.

Screenshots

Ask MiCA — landing. Clean entry point with example prompts and the Citations panel (warm-paper + EU-blue design system, no Tailwind).

RETRIEVAL-AUGMENTED · AGENTIC · CITED

Ask MiCA

A copilot for the EU Markets in Crypto-Assets Regulation (Regulation (EU) 2023/1114). Every answer is grounded in retrieved regulation text and cited by article; the agent can also look up the ESMA register snapshot.

Ask a question about the EU Markets in Crypto-Assets Regulation. Answers are grounded in the indexed regulation text and cited by article.

Ask

CITATIONS

Documents and news cited by the answer appear here.

Ask MiCA — a grounded answer. The agent's tool calls show as chips, the answer streams in, and the Citations panel lists the cited provisions (article/instrument + link).

MICA Copilot

EU crypto-regulation, grounded & cited

Ask MICA

Classify

RETRIEVAL-AUGMENTED - AGENTIC - CITED

Ask MICA

A copilot for the EU Markets in Crypto-Assets Regulation (Regulation (EU) 2023/1141). Every answer is grounded in retrieved regulation text and cited by article; the agent can also look up the ESMA register snapshot.

What are the reserve requirements for an asset-referenced token issuer?

• Searched MICA corpus for "reserve requirements asset-referenced token issuer" + 6 provisions(s)

• Searched MICA corpus for "investment policy reserve assets asset-referenced token Article 36" + 8 provisions(s)

Here is a comprehensive overview of the reserve requirements for **asset-referenced token (ART) issuers** under MICA, drawn from Articles 35–39 of the Regulation and the EBA's draft RTS under Article 36(4).

1. Obligation to Maintain a Reserve [Article 36]

An ART issuer must **"constitute and at all times maintain a reserve of assets"**. The reserve must be composed and managed so that:

- **"(a)"** The risks associated with the referenced assets are covered; and
- **"(b)"** Liquidity risks arising from holders' **"permanent right of redemption"** are addressed.

The aggregate value of the reserve (calculated at market prices) must **"at all times be at least equal"** to the aggregate value of the outstanding claims of token holders [Article 36(7)].

Issuances and redemptions must always be matched by a **"corresponding increase or decrease"** in the reserve [Article 36(6)].

2. Segregation [Article 36(2)-(3) & Article 36(5)]

- The reserve must be **"legally segregated"** from the issuer's own estate — creditors of the issuer have **"no recourse"** to the reserve, including in insolvency.
- The reserve must be **"operationally segregated"** from the issuer's estate and from any other token's reserve.
- Issuers offering **"two or more ARTs"** must maintain **"separate, independently managed pools"** for each token.

3. Own Funds Requirements [Article 35]

The issuer must hold own funds equal to **"the highest"** of:

- **"(a)"** EUR 350,000;
- **"(b)"** 2% of the average amount of the reserve of assets (calculated over the preceding 6 months); or
- **"(c)"** One quarter of fixed overheads of the preceding year.

Own funds must consist of **"Common Equity Tier 1"** items under Regulation (EU) No 575/2013 [Article 35(2)].

The competent authority may require **"up to 20% additional"** own funds where risk factors such as reserve quality, volatility, or investment policy warrant it [Article 35(3)], and **"20%-40% more"** based on stress-testing outcomes [Article 35(5)].

4. Custody of Reserve Assets [Article 37]

Reserve assets must be held in custody **"within 5 working days"** of issuance by:

- A **"CASP"** (for crypto-asset reserve assets),
- A **"credit institution"** (for all asset types), or
- An **"investment firm"** (for financial instrument reserve assets).

Key requirements:

- Reserve assets must **"not be encumbered or pledged"**.
- The issuer must have **"prompt access"** to meet redemption requests.
- **"Concentration risk"** among custodians must be avoided.
- The custodian must be a **"separate legal person"** from the issuer [Article 37(4)].

5. Investment of Reserve Assets [Article 38]

If a portion of the reserve is invested, it must only be placed in **"highly liquid financial instruments"** with **"minimal market, credit, and concentration risk"**, capable of being liquidated rapidly with minimal price impact.

- **"UCITS funds"** are deemed compliant where they invest solely in assets specified by EBA [Article 38(2)].
- All profits, losses, and counterparty/operational risks from investments **"are borne by the issuer"** — not token holders [Article 38(4)].
- EBA is mandated to issue RTS specifying eligible highly liquid instruments and concentration limits [Article 38(5)].

6. Liquidity Requirements (EBA RTS under Article 36(4))

The EBA's draft RTS adds granular liquidity rules, including:

- A required **"percentage of reserve assets with daily maturities"** (redeemable within 1 working day).
- A required **"percentage with weekly maturities"** (redeemable within 5 working days).
- A **"minimum of 30% of the amount in each referenced official currency"** to be held as deposits at credit institutions (rising to **"60% for significant ARTs"**).
- Overall **"liquidity management techniques"** and creditworthiness standards for deposit counterparties.

7. Stabilisation Policy & Audit [Article 36(8)-(10)]

Issuers must publish a **"detailed stabilisation policy"** covering: composition of the reserve, risk assessment, issuance/redemption procedures, and investment policy.

An **"independent audit"** of the reserve must be conducted **"every six months"**, with results notified to the competent authority within 6 weeks and published within 2 weeks thereafter.

8. Redemption Rights [Article 39]

Holders have a **"permanent right of redemption"** (at any time, at no fee) — either in cash (equivalent to market value) or via delivery of the referenced assets. The issuer must maintain clear redemption policies covering valuation, timeframes, and stressed-market procedures.

9. Recovery Plan [Article 46]

Issuers must draw up a **"recovery plan"** (notified to the competent authority within 6 months of authorisation) covering measures if reserve requirements are breached, including liquidity fees on redemptions, daily limits on redemptions, or suspension of redemptions.

This is general information about MICA, not legal advice.

Ask about MICA... Ask

CITATIONS

Regulation & documents

Article 36 ⁹ [regulation](#)

Obligation to have a reserve of assets, and composition and management of such reserve of assets

1. Issuers of asset-referenced tokens shall constitute and at all times maintain a reserve of assets. The reserve of assets shall be composed and managed in such a way that: (a) the risks associated to the assets refer...

Article 35 ⁹ [regulation](#)

Own funds requirements

1. Issuers of asset-referenced tokens shall, at all times, have own funds equal to an amount of at least the highest of the following: (a) EUR 350 000; (b) 2 % of the average amount of the reserves of assets referred to...

Article 37 ⁹ [regulation](#)

Custody of reserve assets

1. Issuers of asset-referenced tokens shall establish, maintain and implement custody policies, procedures and contractual arrangements that ensure at all times that: (a) the reserve assets are not encumbered nor pledg...

EBA its: EBA Consultation Paper - Draft RTS to further sp ⁹ [reg](#)

EBA Consultation Paper - Draft RTS to further specify the liquidity requirements of the reserve of assets under Article 36(4) of MICA's Regulation (EU) 2023/1141

4.2. Executive Summary Article 36(4) of MICA requires issuers of asset-referenced tokens, either if the asset-referenced tokens are classified as significant or not, to constitute and at all times maintain a reser...

Article 48 ⁹ [regulation](#)

Recovery plan

1. An issuer of an asset-referenced token shall draw up and maintain a recovery plan providing for measures to be taken by the issuer to restore compliance with the requirements applicable to the reserve of assets in L...

Article 39 ⁹ [regulation](#)

Right of redemption

1. Holders of asset-referenced tokens shall have a right of redemption at all times against the issuer of the asset-referenced tokens, and in respect of the reserve assets when issuers are not able to meet their oblig...

Article 38 ⁹ [regulation](#)

Investment of the reserve of assets

1. Issuers of asset-referenced tokens that invest a part of the reserve of assets shall only invest those assets in highly liquid financial instruments with minimal market risk, credit risk and concentration risk. The...

Classify — structured result. A euro stablecoin is classified as an **e-money token (EMT)** with the implied services, the key obligations (each tied to an article), and citations — rendered from the JSON-schema-constrained `/classify` response.

STRUCTURED OUTPUT · JSON SCHEMA

Classify under MiCA

Describe a token or service and the copilot classifies it (ART / EMT / other crypto-asset / out of scope), flags the crypto-asset services involved, and lists the key obligations — grounded in retrieved provisions and returned as a structured result.

A token pegged 1:1 to the euro, redeemable at par on demand, backed by euro bank deposits and short-term EU government bonds.

Classify under MiCA

A token pegged 1:1 to the euro, redeemable at par on...

A token whose value is stabilised by referencing a b...

A platform that lets users swap one crypto-asset for...

e-money token (EMT)

confidence: high

The token is pegged 1:1 to the euro (an official EU currency), redeemable at par on demand, and backed by euro-denominated assets (bank deposits and short-term government bonds). This squarely matches the EMT definition and the issuance/redeemability rules in Article 49 MiCA: issuers must issue EMTs at par value and redeem them at any time at par value upon request [Article 49(3)–(4)]. The backing by at least 30% in credit institution deposits and the remainder in secure, low-risk, highly liquid instruments denominated in the same official currency is precisely the investment framework mandated by Article 54 for EMT reserve funds. The token is therefore an e-money token under Title IV of MiCA.

Key obligations

Issuer must be authorised as a credit institution or electronic money institution (EMI) before offering the EMT to the public or seeking its admission to trading. [Article 48(1)(a)]

Issuer must notify a crypto-asset white paper to the competent authority and publish it, at least 40 working days before the intended offer or admission to trading. The white paper must prominently state the conditions for redemption. [Article 48(1)(b), 48(6), Article 49(5), Article 51]

EMTs must be issued at par value upon receipt of funds, and redeemed at par value at any time upon request by the holder, with no fee charged for redemption. [Article 49(3), 49(4), 49(6)]

At least 30% of funds received in exchange for EMTs must be deposited in separate accounts at credit institutions; the remainder must be invested in secure, low-risk, highly liquid assets denominated in euro (the referenced currency). [Article 54(a)-(b)]

Holders of EMTs have a legal claim against the issuer. [Article 49(2)]

Issuer must comply with reporting obligations using the prescribed templates, including data on token type, referenced currency, and whether the token has been classified as significant. [IR 2024/2902 Art. 1; Article 22(1) of Regulation (EU) 2023/1114]

Citations

Article 48 ↗ — Requirements for the offer to the public or admission to trading of e-money tokens

Article 49 ↗ — Issuance and redeemability of e-money tokens

Article 54 ↗ — Investment of funds received in exchange for e-money tokens

IR 2024/2902 Art. 1 ↗ — General provisions — Commission Implementing Regulation (EU) 2024/2902 on reporting templates

ESMA guideline: Final Report - Guidelines on the conditions and ↗ — Final Report - Guidelines on the conditions and criteria for the qualification of crypto-assets as financial instruments (ESMA75453128700-1323)

Article 39 ↗ — Right of redemption

7. GenAI techniques (detail)

- **Multi-source RAG** — two pgvector corpora: a **document corpus** (`reg_chunks` : the Regulation + RTS/ITS + ESMA/EBA guidelines/Q&As, article/section chunked) and a **full-text news corpus** (`news_chunks` ,

recency-weighted). Answers must cite retrieved sources; the prompt forbids answering from general knowledge and abstains when support is missing.

- **Tool / function calling + routing** — the agent picks `search_regulation` (what the law requires), `search_news` (current facts, dated), `lookup_register`, or `check_enforcement`; each tool's description states *when* to call it. News is flagged time-sensitive and cited with its date.
- **Structured outputs** — `/classify` constrains the response to a JSON schema (`output_config.format`), so the UI always renders valid structured data.
- **Prompt engineering / system prompts** — a role-based prompt ships a stable MiCA "map" and an explicit abstention rule.
- **Prompt caching** — the large stable system prefix carries `cache_control`, so repeated requests pay cache-read (~0.1×) instead of full price. *Verified:* a repeat request read ~1,693 tokens from cache (`cache_read_input_tokens`) on the second call.

8. Evaluation results

Run `python -m eval.run --e2e --judge` (golden set in `eval/goldens.jsonl`). Metrics:

Metric	What it measures
<code>retrieval_hit@k</code>	Did the expected article appear among retrieved provisions?
<code>citation_hit</code>	Did the agent's answer actually cite the expected article?
<code>abstention_accuracy</code>	Did the agent decline on out-of-corpus questions?
<code>faithfulness</code>	LLM-as-judge: is every claim supported by retrieved context?

Measured run (13 goldens; agent = Sonnet 4.6, judge = Haiku 4.5, embedder = local `mxbai-embed-large-v1`; **v2 935-chunk document corpus**):

Metric	Score
<code>retrieval_hit@k</code>	0.727
<code>citation_hit</code>	0.818
<code>abstention_accuracy</code>	1.00
<code>faithfulness</code>	0.818

Each golden expects one specific *regulation* article. Against the v2 corpus (which grew from 48 summaries to **935 chunks of real full text** spanning the Regulation **and** the Level-2 RTS/ITS and ESMA/EBA guidelines), that exact article now competes with closely-related Level-2 provisions, so the strict exact-article hit is a little lower than against the old narrow summary corpus — while the answers are *richer* (they cite the RTS **and** the article). Abstention stays perfect and faithfulness holds. Qualitatively, the agent correctly: cited **Article 143** and the **1 July 2026** transitional deadline for the Binance query; identified **Commission Delegated Regulation (EU) 2025/294** as the complaints-handling RTS; and grounded a "recent enforcement" answer in dated news citations. (Full per-question detail: `eval/results/scorecard.json`.)

9. Limitations & future extensions

Limitations. The document corpus is real full text of public EU sources (regulation + RTS/ITS + ESMA/EBA guidelines/Q&As), rebuildable offline from the committed cache or `docs_ingest --refresh`. The news corpus reflects whatever the feeds carried at the last poll — keep it fresh with the scheduler; trade-press full text is stored for private retrieval and always cited + linked with its date (`NEWS_STORE=summary` switches to transformative summaries). The ESMA registers are the **real** public data synced from ESMA's CSVs; Title II white papers have no token-name field, so each token is read from the white-paper document (best-effort — some links are dead/unreadable, so token-name coverage is partial). General information, **not legal advice**.

Future work. (a) Hybrid retrieval (pgvector + the shipped `tsvector` column) and a Voyage reranker. (b) Multilingual (EL/EN) querying (`intfloat/multilingual-e5-large`). (c) A citation-verification self-check pass. (d) Deploy alongside the dashboard (FastAPI systemd unit + Next on PM2 + nginx + the news scheduler).

10. Install / run / examples

See `../README.md` §2–6 for install/run, and `EXAMPLES.md` for a full set of copy-paste usage examples (every endpoint with request **and** response, the SSE event stream, the UI flows, and the eval run). Quick check after setup:

```
curl -s localhost:8000/health | jq          # chunks_indexed (docs), news_indexed, register_rows > 0
curl -s localhost:8000/chat/sync -H 'content-type: application/json' \
  -d '{"message":"Must an e-money token be redeemable at par?"}' | jq '.answer, .citations[].article_ref'
# Level-2 (RTS) + current-facts (news) routing:
curl -s localhost:8000/chat/sync -H 'content-type: application/json' \
  -d '{"message":"Which Delegated Regulation sets the complaints-handling RTS for CASPs?"}' | jq '.citation
curl -s localhost:8000/chat/sync -H 'content-type: application/json' \
  -d '{"message":"What is going on with Binance and MiCA, and is there a deadline?"}' | jq '.tool_events[.]
curl -s localhost:8000/classify -H 'content-type: application/json' \
  -d '{"description":"A token pegged 1:1 to the euro, redeemable at par, backed by bank deposits."}' | jq '
```

11. Deliverables mapping (assignment checklist)

How this project satisfies each requirement of the *AI for Developers — Building with LLMs* brief:

Requirement	Where it lives
Not a chatbot — AI logic inside a real software architecture	FastAPI backend (routers / services / schemas / agents) + Next.js UI; the LLM is one orchestrated layer, not the whole app (§4–§5).
Functional UI	<code>frontend/</code> — Ask MiCA (streaming chat + citations) and Classify (structured card); §6.
Backend API	FastAPI with auto-generated Swagger at <code>/docs</code> ; endpoints in §5 and <code>EXAMPLES.md</code> .
RAG (required)	Multi-source: <code>reg_chunks</code> (official documents) + <code>news_chunks</code> (full-text news), pgvector cosine search, article/source-cited answers, abstention (§7).
Agents / tool-calling (rewarded)	Manual agentic loop routing <code>search_regulation</code> / <code>search_news</code> / <code>lookup_register</code> / <code>check_enforcement</code> (<code>app/services/llm.py</code> , <code>app/agents/tools.py</code>).
Structured outputs (rewarded)	<code>/classify</code> constrained to a JSON schema via <code>output_config.format</code> (§7).
Prompt engineering + caching	Role-based system prompt with an explicit abstention rule; <code>cache_control</code> on the stable prefix (verified cache reads, §7).
Evaluation	<code>eval/</code> golden set + harness; scorecard in §8 (<code>eval/results/scorecard.json</code>).
GitHub repo	This repository.
Documentation (DOC/PDF)	This file (<code>docs/DOCUMENTATION.md</code>) — export to PDF for submission.
README	<code>../README.md</code> .
Usage examples	<code>EXAMPLES.md</code> .
Demo (optional)	The running UI at <code>localhost:3000</code> ; <code>EXAMPLES.md</code> lists one-click example prompts per path.